

RODRIGO GOMES DE ARAÚJO

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PROFILE

MSc Computer Science student graduating in **July 2027**, interested in **ETF trading, quantitative analysis and the systems that support electronic markets**. Experienced in **Python data analysis, PostgreSQL and systems programming**, with projects spanning an analytical data platform, predictive modelling and a decentralized ledger in Rust. Rigorous, reliable and comfortable turning complex data into automated analysis and clear reports. Based in Porto and available to work in Lisbon.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present (Jul 2027) <i>University of Porto (FCUP)</i>
Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FEUP)</i>

PROJECTS

Portuguese Elections Analytics Platform <i>Advanced Topics in Databases · 18/20</i>	<i>PostgreSQL, PostGIS, Python, SQL, OCaml</i>
- Built an end-to-end analytical platform for Portuguese election results, from Python ETL of official Excel data into PostgreSQL to an interactive visual application. - Designed a star-schema data warehouse for votes, turnout and seats, with spatial data and indexes supporting analysis from national to parish level. - Implemented analytical SQL for trend, vote-swing, turnout and party-performance analysis, using roll-ups, CUBE, window functions and geospatial queries.	

PoW Blockchain for Decentralized Auctions <i>Data Systems Security · 19/20</i>	<i>Rust, Tokio, LibP2P, Docker, Blake2b, Ed25519</i>
- Built a decentralized public ledger for auctions, where signed transactions are validated by network participants and recorded in an immutable chain. - Implemented proof-of-work consensus, transaction validation, Merkle-tree block commitments and Kademlia peer discovery. - Automated multi-node Docker simulations with honest and malicious participants, developing skills in distributed trust, fault resilience and adversarial testing.	

UFC Fight Outcome Prediction System <i>Artificial Intelligence · 20/20</i>	<i>Python, pandas, NumPy, scikit-learn, Streamlit</i>
- Built an end-to-end machine-learning system to predict fight outcomes from historical data, engineering 30+ predictive features from performance, experience and physical statistics. - Trained and evaluated six models. Logistic Regression, Random Forest, Gradient Boosting, SVM, Neural Network and a soft-voting ensemble, using five-fold cross-validation and explicit overfitting checks. - Delivered an interactive Streamlit application with probability estimates, model comparisons, feature analysis and visual reports.	

RESEARCH

CaST: Cryptographically-aware Session Types <i>Master's Thesis</i>	Jul 2026 – Present (Jul 2027) <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
- Researching how to make communication protocols correct and secure by construction, so invalid cryptographic operations are rejected before deployment. - Developing formal models and compiler support for tracking authorization and cryptographic evidence across nested protocols, with applications to financial transaction and smart-contract workflows.	

TECHNICAL SKILLS

Programming & Data: Python, SQL, PostgreSQL, Rust, C/C++, Haskell, OCaml, Java
Data & Systems: pandas, NumPy, scikit-learn, PostGIS, Git, UNIX, Docker, Tokio, LibP2P, Terraform, OpenMP
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
Built a game from scratch in 48 hours using Rust and Raylib.	
Academic Merit Awards	2020, 2021, 2022
1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016